TERTIARY CHW MOTOR STARTER & BMS INTERFACE SCHEMATIC:
1. Three-pump variable speed manifold supplying chilled water to Assembly Hall and Cabinet high-comfort zones.
2. Closed-loop PID controller taking input from two differential pressure sensors placed at the ends of supply loops.
3. Microprocessor VFD drives with harmonic mitigation chokes keeping total harmonic distortion (THDi) < 3%.
4. Hardwired interlock with motorized 2-way modulating control valves on terminal AHUs.
5. Thermal overload and phase angle monitoring triggering instant standby pump start in less than 2.5 seconds.

Pump Unit	Flow Capacity	Head Metres	Motor Rating	Inverter Drive	MCCB Protection
PMP-TER-01 (Duty)	180 m3/hr	32 Metres	22.0 kW IE4	ABB ACH580 VFD	63A 3P 36kA
PMP-TER-02 (Duty)	180 m3/hr	32 Metres	22.0 kW IE4	ABB ACH580 VFD	63A 3P 36kA
PMP-TER-03 (Standby)	180 m3/hr	32 Metres	22.0 kW IE4	ABB ACH580 VFD	63A 3P 36kA

AMARAVATI GOVERNMENT COMPLEX · LEGISLATIVE ASSEMBLY		
DWG TITLE: CHILLED WATER TERTIARY PUMP STARTER & MODULAT		
DRAWING NO: SL-288 _ AMARAVATI ASSEMBLY SC	REV: 1	SCALE: 1:100
CLIENT: AP CRDA / GOVT OF ANDHRA PRADESH	DATED: 28-09-2026	